CONTENTS

1. AE 2026

A. Formulae

1.1.1

$$\Pr(M)+\Pr(E)-\Pr(ME)+\Pr(M+E)'=1 \tag{1.1.1.1}$$

1.1.2

$$A+A'=1,\quad AA'=0 \tag{1.1.2.1}$$

$$\implies \Pr(A+A')=\Pr(1)=1\quad \Pr(AA')=\Pr(0)=0 \tag{1.1.2.2}$$

1.1.3

$$\Pr(A+A')=\Pr(A)+\Pr(A') \tag{1.1.3.1}$$

1.1.4 Difference rule of probability

$$\Pr(AB')=\Pr(A)-\Pr(AB) \tag{1.1.4.1}$$

Solution: From (**??**)

$$A=A\cdot 1=A(B+B')=AB+AB' \tag{1.1.4.2}$$

Also,

$$(AB)(AB')=0 \tag{1.1.4.3}$$

From (**??**),

$$\Pr(A)=\Pr(AB+AB')=\Pr(AB)+\Pr(AB') \tag{1.1.4.4}$$

Rearranging, we get:

$$P(AB')=P(A)-P(AB) \tag{1.1.4.5}$$

1.1.5 The Laplace transform is defined as

$$g(t)\overset{\mathcal{L}}{\longleftrightarrow}G(s)=\int_0^\infty e^{-st}g(t)\,dt \tag{1.1.5.1}$$

1.1.6

$$g'(t)\overset{\mathcal{L}}{\longleftrightarrow}sG(s)-g(0) \tag{1.1.6.1}$$

Solution: Applying integration by parts,

$$u=e^{-st},dv=g'(t)\,dt \tag{1.1.6.2}$$

$$\implies \int_0^\infty e^{-st}g'(t)\,dt=\left[e^{-st}g(t)\right]_0^\infty-\int_0^\infty\left(-se^{-st}\right)g(t)\,dt \tag{1.1.6.3}$$

$$=-g(0)+s\int_0^\infty e^{-st}g(t)\,dt \tag{1.1.6.4}$$

yielding (**??**).

1.1.7

$$g''(t) \overset{\mathcal{L}}{\longleftrightarrow} s^2 G(s) - sg(0) - g'(0) \tag{1.1.7.1}$$

1.1.8

$$e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s-a) \tag{1.1.8.1}$$

Solution:

$$\implies e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} \int_0^\infty e^{-st} \left(e^{at} g(t)\right) dt \tag{1.1.8.2}$$

$$= \int_0^\infty e^{-(s-a)t} g(t)\, dt \tag{1.1.8.3}$$

yielding (**??**).

1.1.9 The unit step function is defined as

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases} \tag{1.1.9.1}$$

1.1.10

$$u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s} \tag{1.1.10.1}$$

Solution:

$$\int_0^\infty e^{-st}\, dt = \left(-\frac{e^{-st}}{s}\right)_0^\infty = \frac{1}{s} \quad \Re(s) > 0 \tag{1.1.10.2}$$

1.1.11

$$tu(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s^2} \tag{1.1.11.1}$$

Solution: Differentiating (**??**) with respect to s results in (**??**).

$$\begin{aligned} \cos(at)u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{s}{s^2+a^2} \\ \sin(at)u(t) &\overset{\mathcal{L}}{\longleftrightarrow} \frac{a}{s^2+a^2} \end{aligned} \quad \Re(s) > a \tag{1.1.11.2}$$

Solution: From (**??**) and (**??**),

$$e^{\jmath at} u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s-\jmath a}, \quad \Re(s) > a \tag{1.1.11.3}$$

$$\implies \cos(at)u(t) + \jmath \sin(at)u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{s+\jmath a}{(s-\jmath a)(s+\jmath a)} \tag{1.1.11.4}$$

$$= \frac{s}{s^2+a^2} + \jmath \frac{a}{s^2+a^2} \tag{1.1.11.5}$$

resulting in (**??**).

1.1.12 Solve

$$\frac{d^2y}{dx^2} = -\alpha^2 y, \alpha \neq 0 \tag{1.1.12.1}$$

Solution: Applying the Laplace transform on both sides, and using (**??**),

$$s^2Y(s) - sy(0) - y'(0) = -\alpha^2 Y(s) \tag{1.1.12.2}$$

Substituting

$$y\left(0\right) = \beta, y'\left(0\right) = \gamma \tag{1.1.12.3}$$

in (**??**),

$$Y(s) = \frac{s\beta + \gamma}{s^2 + \alpha^2} \tag{1.1.12.4}$$

$$= \frac{s\beta}{s^2 + \alpha^2} + \frac{\gamma}{s^2 + \alpha^2} \tag{1.1.12.5}$$

yielding

$$y(t) = \beta\cos(\alpha t) + \frac{\gamma}{\alpha}\sin(\alpha t) \tag{1.1.12.6}$$

from (**??**).

1.1.13 (Euler's Method of Forward Differences.) Numerically solve

$$\frac{dy}{dx} = f\left(x, y\right), y\left(x_0\right) = y_0 \tag{1.1.13.1}$$

Solution: By definition,

$$\left.\frac{dy}{dx}\right|_{x=x_0} = \lim_{h\to 0}\frac{y(x_0+h) - y(x_0)}{h} \tag{1.1.13.2}$$

which can be expressed using the notation

$$y_0' = \frac{y_1 - y_0}{h} \tag{1.1.13.3}$$

Similarly,

$$y_n'' = \frac{y_{n+1}' - y_n'}{h} \tag{1.1.13.4}$$

From (**??**),

$$y_{n+1}' = \frac{y_{n+2} - y_{n+1}}{h}, y_n' = \frac{y_{n+1} - y_n}{h} \tag{1.1.13.5}$$

yielding

$$y_n'' = \frac{y_{n+2} - 2y_{n+1} + y_n}{h^2} \tag{1.1.13.6}$$

upon simplification.

1.1.14 Kronecker delta function

$$\delta_n = \begin{cases} 1, & n = 0 \\ 0, & \text{otherwise} \end{cases} \tag{1.1.14.1}$$

1.1.15

$$f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi n f_0 t}\, dt = \delta_n \tag{1.1.15.1}$$

1.1.16 The Fourier Series. Let

$$x(t) = \sum_{k=-\infty}^{\infty} x_0(t)\,(t - nT_0) \tag{1.1.16.1}$$

be a periodic signal with

$$x(t) = \begin{cases} x_0(t), & t \in \left(-\frac{T_0}{2}, \frac{T_0}{2}\right) \\ 0 & \text{otherwise} \end{cases}. \tag{1.1.16.2}$$

Then,

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{\mathrm{j}2\pi n f_0 t}, \quad f_0 = \frac{1}{T_0} \tag{1.1.16.3}$$

Show that

$$c_n = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt \tag{1.1.16.4}$$

Solution: From (**??**),

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} \left(\sum_{k=-\infty}^{\infty} c_k e^{\mathrm{j}2\pi k f_0 t} \right) e^{-\mathrm{j}2\pi n f_0 t}\, dt \tag{1.1.16.5}$$

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\, e^{-\mathrm{j}2\pi n f_0 t}\, dt = \sum_{k=-\infty}^{\infty} c_k \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} e^{\mathrm{j}2\pi (k-n) f_0 t}\, dt \tag{1.1.16.6}$$

$$= \frac{1}{f_0} \sum_{k=-\infty}^{\infty} c_k \delta_{k-n} \tag{1.1.16.7}$$

upon substituting from (**??**) and (**??**), yielding (**??**).

1.1.17 The rectangular function. The unit rectangular function is defined as

$$\text{rect}\,(t) = \begin{cases} 1 & t \in \left(-\frac{1}{2}, \frac{1}{2}\right) \\ 0 & \text{otherwise} \end{cases} \tag{1.1.17.1}$$

1.1.18 Square Wave. Consider the Fourier series with

$$x_0(t) = \text{rect}\,(t), \quad T_0 = 2. \tag{1.1.18.1}$$

Show that

$$c_n = \frac{\sin\left(\frac{\pi k}{2}\right)}{\pi k} \tag{1.1.18.2}$$

Solution: Substituting $f_0 = \frac{1}{2}$ in (**??**),

$$c_n = \frac{1}{2}\int_{-1}^{1} e^{-j\pi kt}\,dt \tag{1.1.18.3}$$

$$= \left[\frac{\sin(\pi kt)}{2\pi k}\right]_{-\frac{1}{2}}^{\frac{1}{2}} \tag{1.1.18.4}$$

The following code verifies the result in Fig. **??**. Note that Fig. **??** is continuous while (**??**) is discontinuous. So (**??**) provides only an approximation. This is known as the Gibbs' phenomenon.

codes/Fourier.py

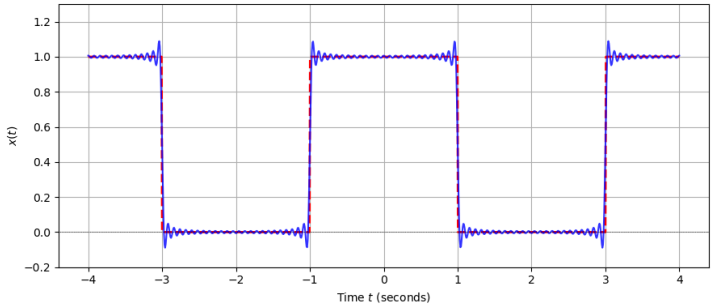


Fig. 1.1.18.1. Gibbs phenomenon

1.1.19 If

$$x(t) \overset{\mathcal{F}}{\longleftrightarrow} c_n, \tag{1.1.19.1}$$

show that

$$x(t-t_0) \overset{\mathcal{F}}{\longleftrightarrow} c_n e^{-\mathrm{J}2\pi nt_0}, \tag{1.1.19.2}$$

Solution:

$$\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t-t_0)\,e^{-\mathrm{J}2\pi nf_0t}\,dt = \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\,e^{-\mathrm{J}2\pi nf_0(t+t_0)}\,dt \tag{1.1.19.3}$$

$$= e^{\mathrm{J}2\pi nf_0t_0}\int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x_0(t)\,e^{-\mathrm{J}2\pi nf_0t}\,dt \tag{1.1.19.4}$$

after a change of variables. Note that the limits of integration are unchanged due the periodicity of $x(t)$. Transferring the exponential term in (**??**) to the L.H.S yields (**??**).

1.1.20 The standard 2-D basis vectors are defined as

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{1.1.20.1}$$

Also,

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{1.1.20.2}$$

1.1.21 For

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top, \text{rank}\,(A) = 1. \tag{1.1.21.1}$$

where both u and v are n-dimensional real-valued non-zero column vectors.
Solution: Let

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}. \tag{1.1.21.2}$$

Then

$$\mathbf{A} = \mathbf{u}\mathbf{v}^\top = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 4 & 2 \end{pmatrix} \tag{1.1.21.3}$$

$$= \begin{pmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 3 & 12 & 6 \end{pmatrix} \tag{1.1.21.4}$$

Following row reduction,

$$\mathbf{A} \xrightarrow{R_2 \to R_2 - 2R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 3 & 12 & 6 \end{pmatrix} \xrightarrow{R_3 \to R_3 - 3R_1} \begin{pmatrix} 1 & 4 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{1.1.21.5}$$

yielding (**??**).

1.1.22

$$\left|\mathbf{A}^\top\right| = \left|\mathbf{A}\right| \tag{1.1.22.1}$$

1.1.23

$$\left|\mathbf{A}\mathbf{B}\right| = \left|\mathbf{A}\right| \left|\mathbf{B}\right| \tag{1.1.23.1}$$

Solution: If $\mathbf{A}$ is invertible

$$\mathbf{A} = \mathbf{E}_1 \mathbf{E}_2 \cdots \mathbf{E}_k \tag{1.1.23.2}$$

where $\mathbf{E}_i$ are invertible matrices. For any single elementary matrix $\mathbf{E}$, it is easy to

verify that

$$\left|\mathbf{EB}\right| = \left|\mathbf{E}\right|\left|\mathbf{B}\right|. \tag{1.1.23.3}$$

$$\therefore \left|\mathbf{AB}\right| = \left|\mathbf{E}_1\mathbf{E}_2\cdots\mathbf{E}_k\mathbf{B}\right| \tag{1.1.23.4}$$

$$= \left|\mathbf{E}_1\right|\left|\mathbf{E}_2\cdots\mathbf{E}_k\mathbf{B}\right| \tag{1.1.23.5}$$

$$= \left|\mathbf{E}_1\right|\left|\mathbf{E}_2\right|\cdots\left|\mathbf{E}_k\right|\left|\mathbf{B}\right| \tag{1.1.23.6}$$

For $\mathbf{B} = \mathbf{I}$,

$$\left|\mathbf{A}\right| = \left|\mathbf{E}_1\right|\left|\mathbf{E}_2\right|\cdots\left|\mathbf{E}_k\right| \tag{1.1.23.7}$$

which upon substituting in (**??**) yields

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\right| \tag{1.1.23.8}$$

1.1.24

$$\left|\mathbf{A^{-1}}\right| = \frac{1}{\left|\mathbf{A}\right|} \tag{1.1.24.1}$$

1.1.25 If

$$\mathbf{Ax} = \lambda\mathbf{x}, \tag{1.1.25.1}$$

then λ is an eigenvalue $\mathbf{x}$ of $\mathbf{A}$.

1.1.26 Let $\lambda_1, \lambda_2, ..., \lambda_i$ be the eigenvalues of a matrix $\mathbf{A}$. Then,

$$\sum_{i=1}^{n}\lambda_i = \mathrm{Tr}\left(\mathbf{A}\right) \tag{1.1.26.1}$$

$$\prod_{i=1}^{n}\lambda_i = \left|\mathbf{A}\right| \tag{1.1.26.2}$$

1.1.27 $\mathbf{A}$ and $\mathbf{A}^\top$ have the same eigenvalues.
Solution: From (**??**),

$$\left|\mathbf{A} - \lambda\mathbf{I}\right| = \left|\left(\mathbf{A} - \lambda\mathbf{I}\right)^\top\right| \tag{1.1.27.1}$$

$$= \left|\mathbf{A}^\top - \lambda\mathbf{I}\right|, \tag{1.1.27.2}$$

completing the proof.

1.1.28 The equation of a conic with directrix $\mathbf{n}^\top\mathbf{x} = c$, eccentricity e and focus $\mathbf{F}$ is given by

$$\mathrm{g}\left(\mathbf{x}\right) = \mathbf{x}^\top\mathbf{V}\mathbf{x} + 2\mathbf{u}^\top\mathbf{x} + f = 0 \tag{1.1.28.1}$$

where

$$\mathbf{V} = \left\|\mathbf{n}\right\|^2\mathbf{I} - e^2\mathbf{n}\mathbf{n}^\top, \tag{1.1.28.2}$$

$$\mathbf{u} = ce^2\mathbf{n} - \left\|\mathbf{n}\right\|^2\mathbf{F}, \tag{1.1.28.3}$$

$$f = \left\|\mathbf{n}\right\|^2\left\|\mathbf{F}\right\|^2 - c^2e^2 \tag{1.1.28.4}$$

Solution:

$$\|\mathbf{x}-\mathbf{F}\|^2 = e^2\frac{(\mathbf{n}^\top\mathbf{x}-c)^2}{\|\mathbf{n}\|^2} \tag{1.1.28.5}$$

$$\implies \|\mathbf{n}\|^2(\mathbf{x}-\mathbf{F})^\top(\mathbf{x}-\mathbf{F}) = e^2\left(\mathbf{n}^\top\mathbf{x}-c\right)^2 \tag{1.1.28.6}$$

$$\implies \|\mathbf{n}\|^2\left(\mathbf{x}^\top\mathbf{x}-2\mathbf{F}^\top\mathbf{x}+\|\mathbf{F}\|^2\right) = e^2\left(c^2+\left(\mathbf{n}^\top\mathbf{x}\right)^2-2c\mathbf{n}^\top\mathbf{x}\right) \tag{1.1.28.7}$$

$$= e^2\left(c^2+\left(\mathbf{x}^\top\mathbf{n}\mathbf{n}^\top\mathbf{x}\right)-2c\mathbf{n}^\top\mathbf{x}\right) \tag{1.1.28.8}$$

which can be expressed as (**??**) after simplification.

1.1.29 The polar form of a conic section is given as:

$$r = \frac{ep}{1+e\cos(\theta-\theta_0)} \tag{1.1.29.1}$$

where

- r is the distance from the focus(origin) to a point on the curve.
- e is the eccentricity of the conic
- p is the distance from the focus to the directrix
- θ_0 is the angle that defines the orientation of the major axis of the conic section relative to the polar reference axis ($\theta = 0$).

Solution: From (**??**), the general equation of a conic in matrix form is given by

$$\mathbf{x}^\top\mathbf{V}\mathbf{x}+2\mathbf{u}^\top\mathbf{x}+f=0 \tag{1.1.29.2}$$

For a conic with a focus at the origin ($\mathbf{F}=\mathbf{0}$), unit normal vector $\mathbf{n} = \begin{pmatrix}\cos\theta_0\\ \sin\theta_0\end{pmatrix}$, eccentricity e, and directrix distance parameter $c=-p$, substituting into (**??**)–(**??**) yields:

$$\mathbf{V} = \mathbf{I}-e^2\mathbf{n}\mathbf{n}^\top, \tag{1.1.29.3}$$

$$\mathbf{u} = -pe^2\mathbf{n}, \tag{1.1.29.4}$$

$$f = -p^2e^2 \tag{1.1.29.5}$$

Expressing $\mathbf{x}$ directly in polar coordinates yields

$$\mathbf{x} = r\mathbf{r}(\theta), \quad \text{where } \mathbf{r}(\theta) = \begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix} \tag{1.1.29.6}$$

Substituting into (**??**) yields

$$r^2\left(\mathbf{r}(\theta)^\top\mathbf{V}\mathbf{r}(\theta)\right)+2r\left(\mathbf{u}^\top\mathbf{r}(\theta)\right)+f=0 \tag{1.1.29.7}$$

Evaluating the inner products:

$$\mathbf{n}^\top\mathbf{r}(\theta) = \begin{pmatrix}\cos\theta_0 & \sin\theta_0\end{pmatrix}\begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix} = \cos(\theta-\theta_0) \tag{1.1.29.8}$$

$$\mathbf{r}(\theta)^\top\mathbf{V}\mathbf{r}(\theta) = \mathbf{r}(\theta)^\top\left(\mathbf{I}-e^2\mathbf{n}\mathbf{n}^\top\right)\mathbf{r}(\theta) = 1-e^2\cos^2(\theta-\theta_0) \tag{1.1.29.9}$$

$$\mathbf{u}^\top\mathbf{r}(\theta) = -pe^2\cos(\theta-\theta_0) \tag{1.1.29.10}$$

Substituting these into (**??**):

$$r^2\left[1-e^2\cos^2\left(\theta-\theta_0\right)\right]-2rpe^2\cos\left(\theta-\theta_0\right)-p^2e^2=0 \tag{1.1.29.11}$$

Factoring the quadratic expression:

$$\left[r\left(1+e\cos\left(\theta-\theta_0\right)\right)-ep\right]\left[r\left(1-e\cos\left(\theta-\theta_0\right)\right)+ep\right]=0 \tag{1.1.29.12}$$

Taking the positive real root for radial distance $r>0$:

$$r\left(1+e\cos\left(\theta-\theta_0\right)\right)=ep \tag{1.1.29.13}$$

yielding (**??**)

1.1.30 Gravitational parameter of a body μ is defined as

$$\mu=GM \tag{1.1.30.1}$$

where M is the mass of the body

1.1.31 Specific angular momentum for a satellite with position vector $\mathbf{r}$ and velocity vector $\mathbf{v}$ is

$$h=\|\mathbf{r}\times\mathbf{v}\| \tag{1.1.31.1}$$

1.1.32 Specific Energy for a satellite with position vector $\mathbf{r}$, velocity vector $\mathbf{v}$ and gravitational parameter μ is

$$\varepsilon=\frac{v^2}{2}-\frac{\mu}{r} \tag{1.1.32.1}$$

where

$$r=\|\mathbf{r}\|,v=\|\mathbf{v}\| \tag{1.1.32.2}$$

1.1.33 The gravitational acceleration $\mathbf{a}$ acting on a satellite is given by Newton's law of gravitation:

$$\mathbf{a}=-\frac{\mu}{r^3}\mathbf{r} \tag{1.1.33.1}$$

1.1.34 The trajectory of a satellite under a central gravitational force is always a conic section.

Solution: Taking the cross product of $\mathbf{r}$ with $\mathbf{a}$:

$$\mathbf{r}\times\mathbf{a}=\mathbf{r}\times\left(-\frac{\mu}{r^3}\mathbf{r}\right)=\mathbf{0} \tag{1.1.34.1}$$

From (**??**)

$$\frac{d}{dt}\left(\mathbf{r}\times\mathbf{v}\right)=\mathbf{v}\times\mathbf{v}+\mathbf{r}\times\mathbf{a}=\mathbf{0} \tag{1.1.34.2}$$

Hence, the specific angular momentum vector $\mathbf{h}$ is conserved:

$$\mathbf{h}=\mathbf{r}\times\mathbf{v}=\text{constant} \tag{1.1.34.3}$$

Crossing the acceleration $\mathbf{a}$ with $\mathbf{h}$:

$$\mathbf{a} \times \mathbf{h} = \left(-\frac{\mu}{r^3}\mathbf{r}\right) \times (\mathbf{r} \times \mathbf{v}) \tag{1.1.34.4}$$

$$= -\frac{\mu}{r^3}\left[(\mathbf{r} \cdot \mathbf{v})\,\mathbf{r} - (\mathbf{r} \cdot \mathbf{r})\,\mathbf{v}\right] \tag{1.1.34.5}$$

$$= \frac{\mu}{r^3}\left[r^2\mathbf{v} - (\mathbf{r} \cdot \mathbf{v})\,\mathbf{r}\right] \tag{1.1.34.6}$$

Using $\mathbf{r} \cdot \mathbf{v} = r\frac{dr}{dt}$, this simplifies to:

$$\mathbf{a} \times \mathbf{h} = \mu\left[\frac{\mathbf{v}}{r} - \frac{1}{r^2}\frac{dr}{dt}\mathbf{r}\right] = \mu\frac{d}{dt}\left(\frac{\mathbf{r}}{r}\right) \tag{1.1.34.7}$$

Integrating with respect to time t:

$$\mathbf{v} \times \mathbf{h} = \mu\left(\frac{\mathbf{r}}{r} + \mathbf{C}\right) \tag{1.1.34.8}$$

where $\mathbf{C}$ is the integration constant. Taking the scalar product of (**??**) with $\mathbf{r}$:

$$\mathbf{r} \cdot (\mathbf{v} \times \mathbf{h}) = \mu\mathbf{r} \cdot \left(\frac{\mathbf{r}}{r} + \mathbf{C}\right) \tag{1.1.34.9}$$

$$\implies (\mathbf{r} \times \mathbf{v}) \cdot \mathbf{h} = \mu\left(r + rC\cos\theta\right) \tag{1.1.34.10}$$

$$\implies \|\mathbf{h}\|^2 = \mu r\left(1 + C\cos\theta\right) \tag{1.1.34.11}$$

where θ is the angle between $\mathbf{r}$ and $\mathbf{C}$. Rearranging for r yields

$$r = \frac{\frac{\|\mathbf{h}\|^2}{\mu}}{1 + C\cos\theta} = \frac{ep}{1 + e\cos\theta} \tag{1.1.34.12}$$

This is the same equation as (**??**) where

$$ep = \frac{\|\mathbf{h}\|^2}{\mu} \tag{1.1.34.13}$$

$$\mathbf{e} = \mathbf{C} \tag{1.1.34.14}$$

1.1.35 The squared eccentricity e^2 of a satellite's trajectory is related to its specific orbital energy ε and specific angular momentum magnitude h by

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \tag{1.1.35.1}$$

Solution: From (**??**) and (**??**),

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r} \tag{1.1.35.2}$$

Taking the magnitude squared of both sides using $\|\mathbf{e}\|^2 = \mathbf{e} \cdot \mathbf{e}$:

$$e^2 = \|\mathbf{e}\|^2 = \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}\right) \cdot \left(\frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}\right) \tag{1.1.35.3}$$

Expanding the inner product:

$$e^2 = \frac{\|\mathbf{v}\times\mathbf{h}\|^2}{\mu^2} - \frac{2}{\mu r}\left[\mathbf{r}\cdot(\mathbf{v}\times\mathbf{h})\right] + \frac{\mathbf{r}\cdot\mathbf{r}}{r^2} \tag{1.1.35.4}$$

We know that

$$\mathbf{r}\cdot(\mathbf{v}\times\mathbf{h}) = (\mathbf{r}\times\mathbf{v})\cdot\mathbf{h} = \mathbf{h}\cdot\mathbf{h} = h^2 \tag{1.1.35.5}$$

And,

$$\mathbf{r}\cdot\mathbf{r} = r^2 \tag{1.1.35.6}$$

Substituting these back into (**??**) yields

$$e^2 = \frac{v^2h^2}{\mu^2} - \frac{2h^2}{\mu r} + 1 \tag{1.1.35.7}$$

$$= 1 + \frac{2h^2}{\mu^2}\left(\frac{v^2}{2} - \frac{\mu}{r}\right) \tag{1.1.35.8}$$

Substituting (**??**) into (**??**) yields:

$$e^2 = 1 + \frac{2\varepsilon h^2}{\mu^2} \tag{1.1.35.9}$$

1.1.36 Consider the system A system comprising a bar, spring and mass is shown in the figure Fig. **??** The bar, having negligible mass, is made of a material having Young's modulus E, cross-sectional area A and length L. The spring stiffness is k and the mass is M kg. Find the axial stiffness of the bar.

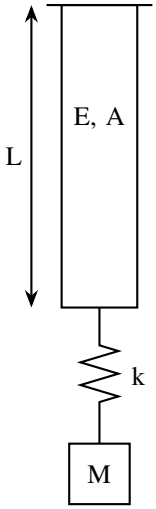


Fig. 1.1.36.1. Spring mass system

Solution: The axial stiffness of the bar is

$$k_b = \frac{AE}{L} \tag{1.1.36.1}$$

1.1.37 Find the overall spring constant in Fig. **??**.

Solution: Using Hooke's law, by balancing the displacements,

$$\frac{Mg}{k_t} = \frac{Mg}{k_b} + \frac{Mg}{k} \tag{1.1.37.1}$$

$$\implies k_{eq} = \frac{k_b k}{k_b + k} \tag{1.1.37.2}$$

where k_t is the overall spring constant.

1.1.38 Show that the equation for motion for the mass in Fig. **??** is

$$\frac{d^2x}{dt^2} = -\frac{k}{M}x \tag{1.1.38.1}$$

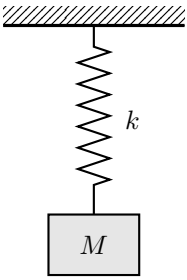


Fig. 1.1.38.1. Spring Mass system in vertical plane

Solution: When the mass M is at rest, let the system stretch by δ_{st}.Then,

$$Mg - k\delta_{st} = 0 \implies Mg = k\delta_{st} \tag{1.1.38.2}$$

When the mass is displaced downward by $x(t)$, using Hooke's law,

$$M\frac{d^2x}{dt^2} = Mg - k(\delta_{\text{st}} + x) \tag{1.1.38.3}$$

$$\tag{1.1.38.4}$$

yielding (**??**), upon substituting from (**??**).

2. BT 2025

A. Formulae

2.1.1 Exponential Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \tag{2.1.1.1}$$

2.1.2 The normal form of the equation of a line is

$$\mathbf{n}^\top \mathbf{x} = c \tag{2.1.2.1}$$

2.1.3 The parametric form of the equation of a line is

$$\mathbf{x} = \mathbf{h} + \kappa \mathbf{m} \tag{2.1.3.1}$$

where

$$\mathbf{m}^\top \mathbf{n} = 0 \tag{2.1.3.2}$$

2.1.4 A circle with center $\mathbf{c}$ and radius r, can be expressed as (**??**) with

$$\mathbf{V} = \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{u} = -\mathbf{c}, \quad f = \|\mathbf{c}\|^2 - r^2 \tag{2.1.4.1}$$

Solution: The equation of a circle is

$$\|\mathbf{x} - \mathbf{c}\|^2 = r^2 \tag{2.1.4.2}$$

$$(\mathbf{x} - \mathbf{c})^\top (\mathbf{x} - \mathbf{c}) = r^2 \tag{2.1.4.3}$$

$$\mathbf{x}^\top \mathbf{x} - 2\mathbf{c}^\top \mathbf{x} + \mathbf{c}^\top \mathbf{c} - r^2 = 0 \tag{2.1.4.4}$$

yielding (**??**) upon comparing with (**??**)

2.1.5 (Line-Conic Intersection) The points of intersection of a line $L : \mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ ($\kappa \in \mathbb{R}$) with a conic section $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ are given by

$$\mathbf{X} = \begin{pmatrix} \mathbf{h} & \mathbf{m} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ \kappa_1 & \kappa_2 \end{pmatrix} \tag{2.1.5.1}$$

where

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) \pm \sqrt{\left[\mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u})\right]^2 - g(\mathbf{h})(\mathbf{m}^\top \mathbf{V}\mathbf{m})}}{\mathbf{m}^\top \mathbf{V}\mathbf{m}}, i = 1, 2. \tag{2.1.5.2}$$

with $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V}\mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$.

2.1.6 (Intersection of Conics and Common Chord) The intersection of two conics with parameters $(\mathbf{V}_1, \mathbf{u}_1, f_1)$ and $(\mathbf{V}_2, \mathbf{u}_2, f_2)$ is given by the pencil

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu \mathbf{V}_2) \mathbf{x} + 2 (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0 \tag{2.1.6.1}$$

The value of μ corresponding to the degenerate conic (the radical axis / common chord line) is found from

$$\begin{vmatrix} \mathbf{V}_1 + \mu \mathbf{V}_2 & \mathbf{u}_1 + \mu \mathbf{u}_2 \\ (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \tag{2.1.6.2}$$

2.1.7 For a differentiable function $f(x)$,

$$f''(x) > 0 \quad \forall x \in (0, \infty) \implies f(x) \text{ is strictly convex} \tag{2.1.7.1}$$

2.1.8 (Gradient Descent Optimization) For the convex optimization problem

$$\min_x f(x), \tag{2.1.8.1}$$

$$x_{k+1} = x_k - \eta f'(x_k) \tag{2.1.8.2}$$

is the iterative parameter update with learning rate $\eta > 0$.

2.1.9 (Newton-Raphson Method) For a real-valued differentiable function $f(x)$, the root is iteratively approximated using its first-order Taylor series expansion about the current estimate x_n

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{2.1.9.1}$$

2.1.10 (System of Linear Equations) The matrix equation

$$\mathbf{A}\mathbf{x} = \mathbf{b} \tag{2.1.10.1}$$

is consistent iff

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \tag{2.1.10.2}$$

2.1.11 Sum of Independent Discrete Random Variables (Two Dice Roll)

- Let $X_1, X_2 \in \{1,2,3,4,5,6\}$ be independent discrete uniform random variables representing the outcomes of two fair dice, each having probability mass function (PMF):

$$p_{X_1}(n) = p_{X_2}(n) = \begin{cases} \frac{1}{6}, & 1 \le n \le 6 \\ 0, & \text{otherwise} \end{cases} \tag{2.1.11.1}$$

- Let $X = X_1 + X_2$ denote their sum. The PMF of X is given by the discrete convolution of $p_{X_1}(n)$ and $p_{X_2}(n)$:

$$p_X(n) = \Pr(X_1 + X_2 = n) = \sum_k \Pr(X_1 = n-k \mid X_2 = k)\, p_{X_2}(k) \tag{2.1.11.2}$$

$$= \sum_k p_{X_1}(n-k)\, p_{X_2}(k) = p_{X_1}(n) * p_{X_2}(n) \tag{2.1.11.3}$$

- Substituting (**??**) into (**??**) yields the triangular PMF

$$p_X(n) = \begin{cases} 0, & n < 2 \\ \frac{n-1}{36}, & 2 \le n \le 7 \\ \frac{13-n}{36}, & 7 < n \le 12 \\ 0, & n > 12 \end{cases} \tag{2.1.11.4}$$

The following code verifies (**??**) in Fig. **??**.

codes/btq35/dice.py

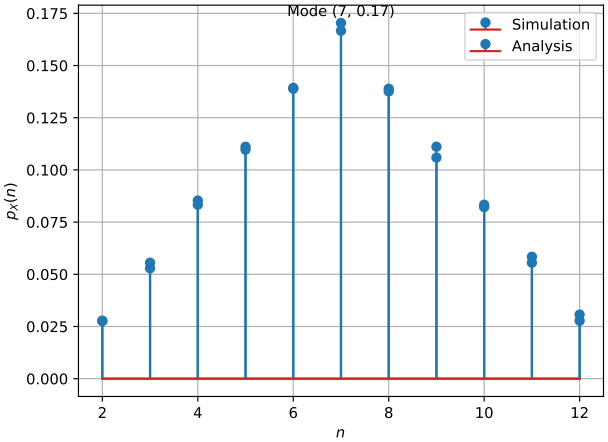


Fig. 2.1.11.1. PMF of the sum of two fair dice: Simulation vs Analysis

2.1.12 (Unit Step Response of an RC Circuit) Consider a series RC circuit excited by a step input $v_i(t) = V_0 u(t)$ where V_0 is a constant.

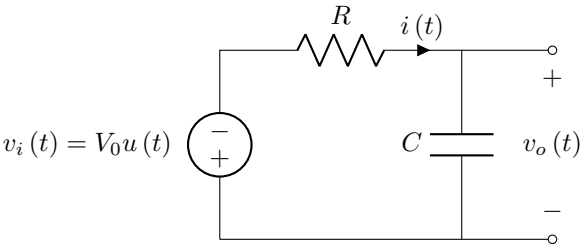


Fig. 2.1.12.1. Time-domain series RC circuit excited by step input $v_i(t)$.

Applying Kirchhoff's Voltage Law (KVL) around the loop:

$$RC\frac{dv_o}{dt} + v_o(t) = V_0 u(t) \tag{2.1.12.1}$$

Transforming the network into the Laplace domain under zero initial conditions ($v_o(0) = 0$), the step voltage source becomes $V_i(s) = \frac{V_0}{s}$, the resistor impedance is R, and the capacitor impedance is $\frac{1}{sC}$.

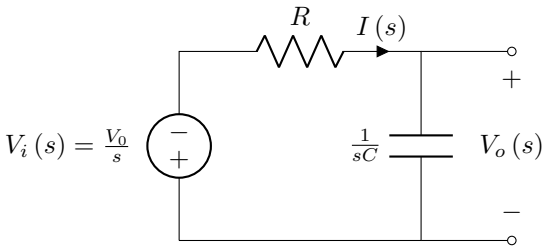


Fig. 2.1.12.2. s-domain equivalent circuit with capacitor impedance $\frac{1}{sC}$.

From the s-domain circuit in Fig. **??**, applying KVL yields:

$$\frac{V_0}{s} = sCRV_o(s) + V_o(s) \tag{2.1.12.2}$$

$$\implies V_o(s) = \frac{V_0}{s(RCs+1)} \tag{2.1.12.3}$$

$$= V_0\left[\frac{1}{s} - \frac{1}{s+\frac{1}{RC}}\right] \tag{2.1.12.4}$$

Taking the inverse Laplace transform gives the output response:

$$v_o(t) = V_0\left(1 - e^{-\frac{t}{RC}}\right)u(t) \tag{2.1.12.5}$$

$$RC\frac{dv_o}{dt} + v_o(t) = V_0 u(t) \tag{2.1.12.6}$$

yielding

$$\frac{V_0}{s} = sCRV_o(s) + V_o(s) \tag{2.1.12.7}$$

$$\implies V_o(s) = \frac{V_0}{s(RCs+1)} \tag{2.1.12.8}$$

$$= V_0\left[\frac{1}{s} - \frac{1}{s+\frac{1}{RC}}\right] \tag{2.1.12.9}$$

$$\implies v_o(t) = V_0\left(1 - e^{-\frac{t}{RC}}\right)u(t) \tag{2.1.12.10}$$

2.1.13 (Continuity) A real-valued function $f(x)$ is continuous at $x = x_0$ iff

$$\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x) = f(x_0) \tag{2.1.13.1}$$

2.1.14 (Differentiability) A function $f(x)$ is differentiable at $x = x_0$ if

$$f'(x_0^-) = \lim_{h\to 0^-}\frac{f(x_0+h)-f(x_0)}{h} = \lim_{h\to 0^+}\frac{f(x_0+h)-f(x_0)}{h} = f'(x_0^+) \tag{2.1.14.1}$$

2.1.15 If $f(x)$ is differentiable at $x = x_0$, it must necessarily be continuous at $x = x_0$

$$\lim_{x \to x_0} [f(x) - f(x_0)] = \lim_{x \to x_0} \left[\frac{f(x) - f(x_0)}{x - x_0} (x - x_0) \right] = f'(x_0) \cdot 0 = 0 \tag{2.1.15.1}$$

2.1.16 Piecewise Boundary Conditions at $x = a$

- Consider a function $f(x)$ defined by differentiable branches $f_1(x)$ and $f_2(x)$ across a boundary point $x = a$:

$$f(x) = \begin{cases} f_1(x), & x \leq a \\ f_2(x), & x > a \end{cases} \tag{2.1.16.1}$$

For $f(x)$ to be continuous at $x = a$, the one-sided limits must equal the boundary value:

$$\lim_{x \to a^-} f_1(x) = \lim_{x \to a^+} f_2(x) \implies f_1(a) = f_2(a) \tag{2.1.16.2}$$

For $f(x)$ to be differentiable at $x = a$, the one-sided derivatives at the boundary must coincide:

$$f_1'(a) = f_2'(a) \tag{2.1.16.3}$$

2.1.17 Solution of First-Order Recurrence Relations via Z-Transform

- The bilateral Z-transform of a discrete-time sequence $x(n)$ is defined as:

$$X(z) = \mathcal{Z}(x(n)) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \tag{2.1.17.1}$$

over the Region of Convergence (ROC) where the series converges absolutely.

- Discrete Unit Step Sequence: The discrete unit step function $u(n)$ is defined as:

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{2.1.17.2}$$

Its forward Z-transform is given by:

$$\mathcal{Z}(u(n)) = \sum_{n=0}^{\infty} z^{-n} = \frac{1}{1 - z^{-1}}, \quad |z| > 1 \tag{2.1.17.3}$$

- Time-Shift Property: Shifting a sequence by an integer delay k corresponds to multiplication by z^{-k} in the Z-domain:

$$\mathcal{Z}(x(n-k)) = z^{-k} X(z) \tag{2.1.17.4}$$

Consequently, delaying the discrete unit step by one sample yields:

$$\mathcal{Z}(u(n-1)) = \frac{z^{-1}}{1 - z^{-1}}, \quad |z| > 1 \tag{2.1.17.5}$$

- Standard Inverse Transform Pair: For any scalar constant α:

$$\mathcal{Z}^{-1}\left(\frac{1}{1-\alpha z^{-1}}\right)=\alpha^{n}u\left(n\right),\quad |z|>|\alpha| \tag{2.1.17.6}$$

2.1.18 (Trace of a Matrix) Let $\mathbf{A}\in\mathbb{R}^{n\times n}$ be a square matrix with eigenvalues $\lambda_1,\lambda_2,\ldots,\lambda_n$. Then,

$$\sum_{i=1}^{n}\lambda_i=\sum_{i=1}^{n}a_{ii}=\operatorname{tr}\left(\mathbf{A}\right) \tag{2.1.18.1}$$

Solution: The characteristic polynomial of $\mathbf{A}$ is defined by

$$p_{\mathbf{A}}\left(\lambda\right)=\left|\lambda\mathbf{I}-\mathbf{A}\right| \tag{2.1.18.2}$$

$$=\prod_{i=1}^{n}\left(\lambda-\lambda_i\right)=\lambda^{n}-\left(\sum_{i=1}^{n}\lambda_i\right)\lambda^{n-1}+\cdots+(-1)^{n}\prod_{i=1}^{n}\lambda_i \tag{2.1.18.3}$$

Evaluating the determinant $\left|\lambda\mathbf{I}-\mathbf{A}\right|$ using cofactor expansion, the only term that contributes to λ^n and λ^{n-1} is the product of the principal diagonal entries

$$\prod_{i=1}^{n}\left(\lambda-a_{ii}\right)=\lambda^{n}-\left(\sum_{i=1}^{n}a_{ii}\right)\lambda^{n-1}+\ldots \tag{2.1.18.4}$$

where a_{ii} denotes the diagonal elements of $\mathbf{A}$. Equating the coefficient of λ^{n-1} from (**??**) and (**??**)

B. Problems

2.2.1 The sum of the following infinite series is:

$$\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}+\frac{1}{5!}+\cdots \tag{2.2.1.1}$$

(BT 2025)

a) π b) $1+e$ c) $e-1$ d) e

Solution: (**??**) can be expressed as

$$\sum_{n=1}^{\infty}\frac{1}{n!}=\sum_{n=0}^{\infty}\frac{1}{n!}-1=e-1 \tag{2.2.1.2}$$

upon substituting from (**??**) The following C code verifies the series sum numerically.

codes/btq3/btq3.c

2.2.2 A circle with center at $(x,y)=(0.5,0)$ and radius $r_1=0.5$ intersects with another circle with center at $(x,y)=(1,1)$ and radius $r_2=1$ at two points. One of the points of intersection (x,y) is: (BT 2025)

a) $(0,0)$ b) $(0.2,0.4)$ c) $(0.5,0.5)$ d) $(1,2)$

Solution: From (**??**), the circle parameters are

$$,\mathbf{c}_1=\begin{pmatrix}0.5\\0\end{pmatrix},r_1=0.5,\quad \mathbf{u}_1=-\mathbf{c}_1=\begin{pmatrix}-0.5\\0\end{pmatrix},\quad f_1=\|\mathbf{c}_1\|^2-r_1^2=0 \tag{2.2.2.1}$$

$$\mathbf{c}_2=\begin{pmatrix}1\\1\end{pmatrix},\ r_2=1,\quad \mathbf{u}_2=-\mathbf{c}_2=\begin{pmatrix}-1\\-1\end{pmatrix},\quad f_2=\|\mathbf{c}_2\|^2-r_2^2=1 \tag{2.2.2.2}$$

Substituting the above in (**??**),

$$\begin{vmatrix}1+\mu & 0 & -0.5-\mu\\ 0 & 1+\mu & -\mu\\ -0.5-\mu & -\mu & \mu\end{vmatrix}=0 \tag{2.2.2.3}$$

$$\implies (1+\mu)\left(-\mu^2-\frac{1}{4}\right)=0 \tag{2.2.2.4}$$

$$\text{or, } \mu=-1 \tag{2.2.2.5}$$

ignoring imaginary roots. Substituting $\mu=-1$ in (**??**),

$$2\left(\mathbf{u}_1-\mathbf{u}_2\right)^{\top}\mathbf{x}+(f_1-f_2)=0 \tag{2.2.2.6}$$

$$\implies \begin{pmatrix}1 & 2\end{pmatrix}\mathbf{x}=1 \tag{2.2.2.7}$$

From (**??**) and (**??**),

$$\mathbf{n} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{m} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}, \mathbf{h} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{2.2.2.8}$$

Substituting the above along with the first circle paramters from (**??**) in (**??**),

$$\mathbf{m}^\top \mathbf{V}_1 \mathbf{m} = \begin{pmatrix} 2 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = 5 \tag{2.2.2.9}$$

$$\mathbf{V}_1 \mathbf{h} + \mathbf{u}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -0.5 \\ 0 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0 \end{pmatrix} \tag{2.2.2.10}$$

$$\mathbf{m}^\top \left(\mathbf{V}_1 \mathbf{h} + \mathbf{u}_1 \right) = \begin{pmatrix} 2 & -1 \end{pmatrix} \begin{pmatrix} 0.5 \\ 0 \end{pmatrix} = 1 \tag{2.2.2.11}$$

$$g\left(\mathbf{h}\right) = \mathbf{h}^\top \mathbf{V}_1 \mathbf{h} + 2\mathbf{u}_1^\top \mathbf{h} + f_1 = 1 - 1 + 0 = 0 \tag{2.2.2.12}$$

yielding

$$\kappa = \frac{-1 \pm \sqrt{1^2 - 0}}{5} \implies \kappa_1 = 0, \kappa_2 = -\frac{2}{5} \tag{2.2.2.13}$$

$\therefore$ the points of intersection are

$$\mathbf{x}_1 = \mathbf{h} + 0\mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{2.2.2.14}$$

$$\mathbf{x}_2 = \mathbf{h} - \frac{2}{5}\mathbf{m} = \begin{pmatrix} 0.2 \\ 0.4 \end{pmatrix} \tag{2.2.2.15}$$

Hence, one of the points of intersection is $(0.2, 0.4)$ The following code verifies this in Fig. **??**.

codes/btq9/circleProblem1.py

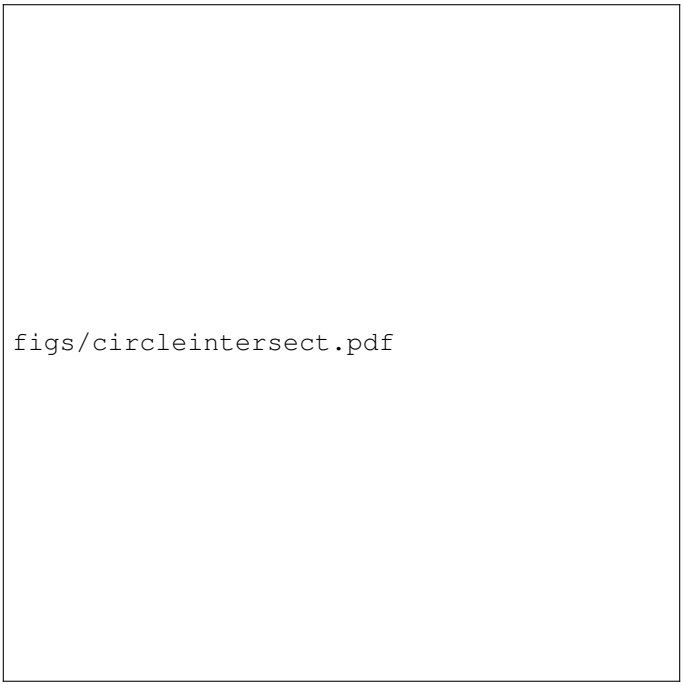


Fig. 2.2.2.1. Intersection of the two circles and the radical axis

2.2.3 Let $y\,(t)$ be a bacterial population whose growth is given by

$$\frac{dy}{dt} = \lambda\,(y+2) \tag{2.2.3.1}$$

where λ is the growth rate constant. If $y\,(0) = 1$ and $y\,(1) = 4$, then the value of λ is (BT 2025)

a) $\ln 2$ b) $\ln 3$ c) $\ln 4$ d) $\ln 6$

Solution: Rearranging the differential equation into standard linear form

$$\frac{dy}{dt} - \lambda y = 2\lambda u(t) \tag{2.2.3.2}$$

Taking the unilateral Laplace transform on both sides using (**??**) and (**??**):

$$sY(s) - y\,(0) - \lambda Y(s) = \frac{2\lambda}{s} \tag{2.2.3.3}$$

Substituting the initial condition $y\left(0\right)=1$,

$$\left(s-\lambda\right)Y\left(s\right)-1=\frac{2\lambda}{s} \tag{2.2.3.4}$$

$$\implies Y\left(s\right)=\frac{s+2\lambda}{s\left(s-\lambda\right)} \tag{2.2.3.5}$$

$$=-\frac{2}{s}+\frac{3}{s-\lambda} \tag{2.2.3.6}$$

Applying the inverse Laplace transform using (**??**) and (**??**),

$$y\left(t\right)=3e^{\lambda t}-2 \tag{2.2.3.7}$$

$$\implies y\left(1\right)=4=3e^{\lambda}-2 \tag{2.2.3.8}$$

$$\text{or, } \lambda=\ln 2 \tag{2.2.3.9}$$

From (**??**), with step size $h=0.01$ and $\lambda=\ln 2$, the following difference equation is obtained.

$$y_{n+1}=y_n+0.01\left[\ln\left(2\right)\left(y_n+2\right)\right],\quad y_0=1 \tag{2.2.3.10}$$

The following codes are used to generate Fig. **??**.

codes/btq13/bac.py

codes/btq13/bac.c

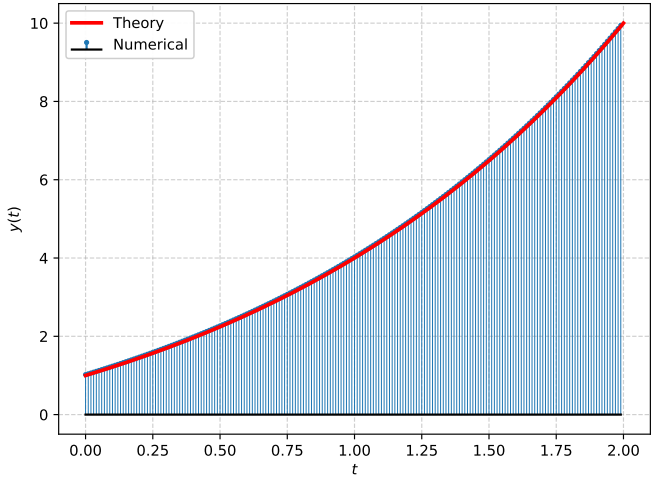


Fig. 2.2.3.1. Euler's Forward Difference Simulation of Bacterial Growth

2.2.4 The minimum value of the function

$$f(x) = x + \frac{4}{x} \tag{2.2.4.1}$$

for $x > 0$ is: (BT 2025)

a) 1 b) 2 c) 3 d) 4

Solution: To locate the critical points of $f(x)$, we set the first derivative to zero:

$$f'(x) = \frac{d}{dx}\left(x + \frac{4}{x}\right) = 1 - \frac{4}{x^2} = 0 \tag{2.2.4.2}$$

$$1 = \frac{4}{x^2} \implies x^2 = 4 \tag{2.2.4.3}$$

Since $x > 0$, the only admissible critical point is:

$$x = 2 \tag{2.2.4.4}$$

From (**??**)

$$f''(x) = \frac{d^2}{dx^2}\left(x + \frac{4}{x}\right) = \frac{8}{x^3} > 0, \quad x \in (0, \infty) \tag{2.2.4.5}$$

Thus, $f(x)$ is strictly convex in the above interval. This is verified by the following code in Fig. **??**.

codes/btq14/min.py

Applying the first-order update relation from (**??**) with learning rate $\eta = 0.3$:

$$x_{k+1} = x_k - 0.3\left(1 - \frac{4}{x_k^2}\right) \tag{2.2.4.6}$$

Starting from the initial guess $x_0 = 5$, the sequence converges to $x^* = 2$, yielding $f(x^*) = 4$. The following code accomplishes this.

codes/btq14/min.c

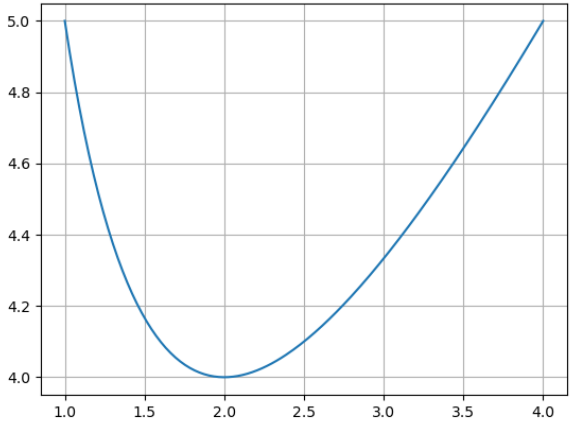


Fig. 2.2.4.1. Plot of $f(x) = x + \frac{4}{x}$ indicating the minimum at $(2, 4)$

2.2.5 Consider a nonlinear algebraic equation, $e^x - 2 = 0$. Using the Newton-Raphson method, with the initial guess of $x_0 = 1$, the approximated value of the root of the equation after one iteration is _____ (round off to two decimal places). (BT 2025)
Solution: Substituting

$$f(x) = e^x - 2 \tag{2.2.5.1}$$
$$f'(x) = e^x \tag{2.2.5.2}$$

into (**??**),

$$x_{n+1} = x_n - \frac{e^{x_n} - 2}{e^{x_n}} \tag{2.2.5.3}$$
$$= x_n - \left(1 - \frac{2}{e^{x_n}}\right) \tag{2.2.5.4}$$
$$= x_n - 1 + 2e^{-x_n} \tag{2.2.5.5}$$

Given the initial guess $x_0 = 1$, the first iteration $(n = 0)$ gives

$$x_1 \approx x_0 - 1 + 2e^{-x_0} \tag{2.2.5.6}$$
$$= 1 - 1 + 2e^{-1} \tag{2.2.5.7}$$
$$= \frac{2}{e} \tag{2.2.5.8}$$

From the following code,

$$x_4 = 0.693147 \approx \ln 2 \tag{2.2.5.9}$$

which is the exact solution.

codes/btq33/raph.c

The following code generates Fig. **??**.

codes/btq33/raph.py

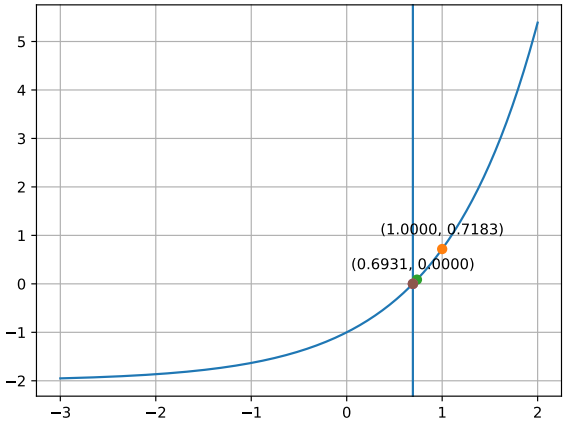


Fig. 2.2.5.1. Convergence of the Newton-Raphson method to $x = \ln 2$

2.2.6 The value of k, for which the linear equations $2x + 3y = 6$ and $4x + 6y = 3k$ have at least one solution, is _____ (answer in integer). (BT 2025)

Solution: Expressing in matrix form $\mathbf{Ax} = \mathbf{b}$,

$$\begin{pmatrix} 2 & 3 \\ 4 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 3k \end{pmatrix} \tag{2.2.6.1}$$

The augmented matrix is

$$(\mathbf{A} \mid \mathbf{b}) = \left(\begin{array}{cc|c} 2 & 3 & 6 \\ 4 & 6 & 3k \end{array}\right) \tag{2.2.6.2}$$

Applying Gaussian elimination using elementary row operations,

$$\left(\begin{array}{cc|c} 2 & 3 & 6 \\ 4 & 6 & 3k \end{array}\right) \xrightarrow{R_2 \to R_2 - 2R_1} \left(\begin{array}{cc|c} 2 & 3 & 6 \\ 0 & 0 & 3k - 12 \end{array}\right) \tag{2.2.6.3}$$

$\because \text{rank}\,(\mathbf{A}) = 1$, from the above, (**??**) is consistent if

$$\text{rank}\left((\mathbf{A} \mid \mathbf{b})\right) = \text{rank}\,(\mathbf{A}) = 1 \tag{2.2.6.4}$$

$$\implies 3k - 12 = 0 \tag{2.2.6.5}$$

$$\text{or, } k = 4 \tag{2.2.6.6}$$

The following code verifies the result by generating Fig. **??**.

codes/btq34/bt34.py

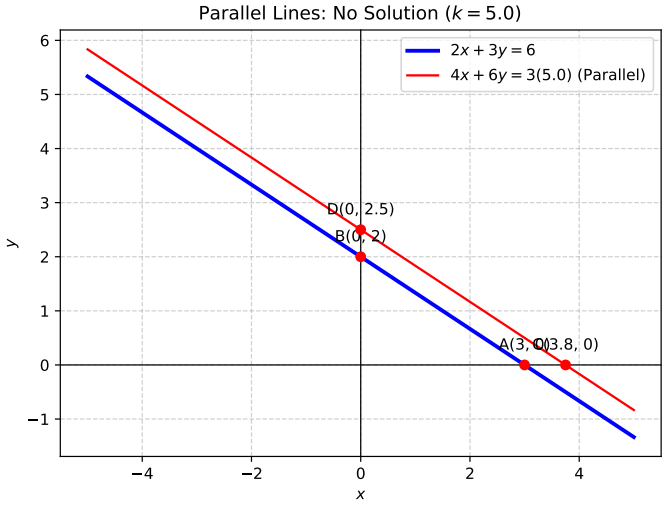


Fig. 2.2.6.1. Coincident lines representing infinitely many solutions at $k = 4$

2.2.7 Two fair six-sided dice, with sides numbered 1 to 6, are thrown once. The probability of getting 7 as the sum of the numbers on the top side of the dice is _____. (BT 2025)

Solution: Let X_1 and X_2 represent the scores on the two dice, and let $X = X_1 + X_2$ denote their sum. From the triangular distribution derived in (**??**),

$$p_X(n) = \frac{n-1}{36}, \quad 2 \le n \le 7 \tag{2.2.7.1}$$

$$\implies \Pr(X = 7) = \frac{1}{6} \tag{2.2.7.2}$$

2.2.8 A thermometer measuring body temperature follows a first-order response with a time constant of 40 seconds. The instrument will reach 95% of its steady-state output at _____ seconds (round off to the nearest integer). (BT 2025)

a) 60 b) 80 c) 120 d) 160

Solution: From (**??**),

$$\frac{v_o(t)}{V_0} = \left(1 - e^{-\frac{t}{RC}}\right) u(t) \tag{2.2.8.1}$$

$$\implies 0.95 = \left(1 - e^{-\frac{t}{40}}\right) \tag{2.2.8.2}$$

$$\implies t = 40 \ln(20) \approx 120 \text{ s} \tag{2.2.8.3}$$

The following code generates Fig. **??**. for verification.

codes/btq42/plot.py

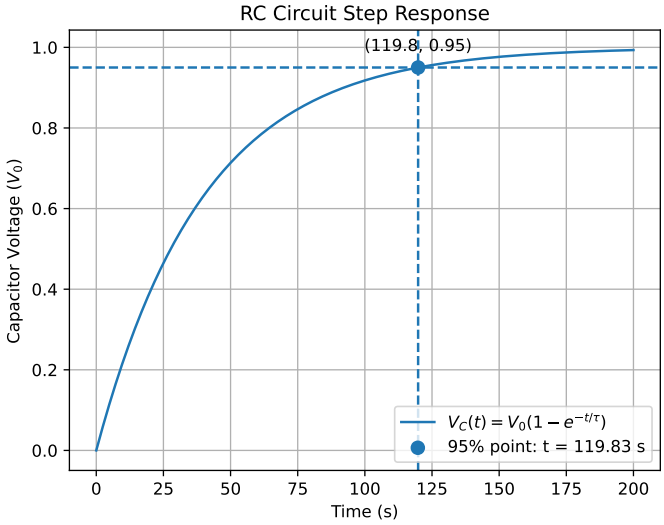


Fig. 2.2.8.1. Step response of the first-order system with $\tau = 40$ s

2.2.9 The output $y(t)$ of a first-order process is governed by the following differential equation

$$\tau_p \frac{dy}{dt} + y = K_p f(t) \tag{2.2.9.1}$$

where τ_p is a non-zero time constant, K_p is the gain and $f(t)$ is the input with $f(0) = 0$. Assume $y(0) = 0$. The transfer function for this process is (consider s as the independent variable in the Laplace domain) (BT 2025)

a) $\frac{K_p}{\tau_p s+1}$ b) $\frac{\tau_p}{K_p s+1}$ c) $\frac{\tau_p}{K_p(s+1)}$ d) $\frac{K_p}{\tau_p(s+1)}$

Solution: In the s-domain, (**??**) becomes

$$\tau_p s Y(s) + Y(s) = K_p F(s) \tag{2.2.9.2}$$

$$\implies \frac{Y(s)}{F(s)} = \frac{K_p}{\tau_p s + 1} \tag{2.2.9.3}$$

2.2.10 Let m and n be fixed real numbers. If the function $y(t) = C_1 e^t + C_2 e^{-t}$ is a solution of

$$\frac{d^2y}{dt^2} + m\frac{dy}{dt} + ny = 0 \tag{2.2.10.1}$$

for any constants C_1 and C_2, then $m + n$ is equal to (BT 2025)

a) -2 b) -1 c) 0 d) 1

Solution: Substituting

$$\frac{dy}{dt} = C_1 e^{t} - C_2 e^{-t} \tag{2.2.10.2}$$

$$\frac{d^2y}{dt^2} = C_1 e^{t} + C_2 e^{-t} \tag{2.2.10.3}$$

Substituting y, $\frac{dy}{dt}$, and $\frac{d^2y}{dt^2}$ into (**??**),

$$\left(C_1 e^{t} + C_2 e^{-t}\right) + m\left(C_1 e^{t} - C_2 e^{-t}\right) + n\left(C_1 e^{t} + C_2 e^{-t}\right) = 0 \tag{2.2.10.4}$$

$$\implies C_1 e^{t}\left(1 + m + n\right) + C_2 e^{-t}\left(1 - m + n\right) = 0 \tag{2.2.10.5}$$

$$\text{or, } m + n = -1 \tag{2.2.10.6}$$

2.2.11 Consider the piecewise function

$$f\left(x\right) = \begin{cases} \sin\left(2x\right), & x > 0 \\ a + bx, & x \leq 0 \end{cases} \tag{2.2.11.1}$$

where a and b are real constants. If $f\left(x\right)$ is differentiable at $x = 0$, then the value of $a + b$ is (BT 2025)

a) 0 b) 1 c) 2 d) 3

Solution: Evaluating the one-sided limits

$$\lim_{x \to 0^+} f\left(x\right) = \lim_{x \to 0^+} \sin\left(2x\right) = \sin\left(0\right) = 0 \tag{2.2.11.2}$$

$$\lim_{x \to 0^-} f\left(x\right) = \lim_{x \to 0^-} \left(a + bx\right) = a + b\left(0\right) = a \tag{2.2.11.3}$$

$$\implies a = 0 \tag{2.2.11.4}$$

from (**??**) and (**??**). Also,

$$f'\left(0^+\right) = 2\cos\left(0\right) = 2 \tag{2.2.11.5}$$

$$f'\left(0^-\right) = b \implies b = 2 \tag{2.2.11.6}$$

from (**??**).

$$\therefore a + b = 2 \tag{2.2.11.7}$$

The following code generates Fig. **??** for verification.

codes/btq46/bt46.py



Fig. 2.2.11.1. Differentiable piecewise function and its tangent line $y = 2x$ at $x = 0$

2.2.12 In a fermentation process, each mole of glucose is converted to biomass ($CH_{1.8}O_{0.5}N_{0.2}$), with a biomass yield coefficient of 0.4 C-mol/C-mol, according to the unbalanced equation given below.

$$C_6H_{12}O_6 + NH_3 + O_2 \rightarrow CH_{1.8}O_{0.5}N_{0.2} + CO_2 + H_2O \tag{2.2.12.1}$$

The moles of oxygen consumption per mole of glucose consumed during fermentation is _____. (BT 2025)

Solution: Let the balanced stoichiometric reaction be represented as

$$x_1C_6H_{12}O_6 + x_2NH_3 + x_3O_2 \rightarrow x_4CH_{1.8}O_{0.5}N_{0.2} + x_5CO_2 + x_6H_2O \tag{2.2.12.2}$$

where $x_1, x_2, x_3, x_4, x_5, x_6$ are the respective stoichiometric coefficients. Formulating the elemental conservation balances for C, N, H, and O using exact rational fractions:

$$\text{Carbon (C):} \quad 6x_1 = x_4 + x_5 \tag{2.2.12.3}$$

$$\text{Nitrogen (N):} \quad x_2 = \frac{1}{5}x_4 \tag{2.2.12.4}$$

$$\text{Hydrogen (H):} \quad 12x_1 + 3x_2 = \frac{9}{5}x_4 + 2x_6 \tag{2.2.12.5}$$

$$\text{Oxygen (O):} \quad 6x_1 + 2x_3 = \frac{1}{2}x_4 + 2x_5 + x_6 \tag{2.2.12.6}$$

Formulating the equivalent matrix equation,

$$\begin{pmatrix} 6 & 0 & 0 \\ 0 & 1 & 0 \\ 12 & 3 & 0 \\ 6 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ \frac{1}{5} & 0 & 0 \\ \frac{9}{5} & 0 & 2 \\ \frac{1}{2} & 2 & 1 \end{pmatrix} \begin{pmatrix} x_4 \\ x_5 \\ x_6 \end{pmatrix} \tag{2.2.12.7}$$

$$\implies \begin{pmatrix} 6 & 0 & 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & -\frac{1}{5} & 0 & 0 \\ 12 & 3 & 0 & -\frac{9}{5} & 0 & -2 \\ 6 & 0 & 2 & -\frac{1}{2} & -2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \tag{2.2.12.8}$$

The biomass yield coefficient is 0.4 C-mol/C-mol $= \frac{2}{5}$ C-mol/C-mol, which provides the fifth relation:

$$x_4 - \frac{2}{5}(6x_1) = 0 \implies \frac{12}{5}x_1 = x_4 \tag{2.2.12.9}$$

Expressing the system for the unknowns $\mathbf{y} = \begin{pmatrix} x_1 & x_2 & x_3 & x_5 & x_6 \end{pmatrix}^\top$ normalized with respect to x_4, we construct the augmented matrix:

$$\begin{pmatrix} \mathbf{A} & \mathbf{b} \end{pmatrix} = \left(\begin{array}{ccccc|c} 6 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{5} \\ 12 & 3 & 0 & 0 & -2 & \frac{9}{5} \\ 6 & 0 & 2 & -2 & -1 & \frac{1}{2} \\ \frac{12}{5} & 0 & 0 & 0 & 0 & 1 \end{array}\right) \tag{2.2.12.10}$$

Applying the sequence of elementary row operations:

- $R_1 \to \frac{1}{6}R_1, \quad R_5 \to \frac{5}{12}R_5$
- $R_3 \to R_3 - 12R_1, \quad R_4 \to R_4 - 6R_1, \quad R_5 \to R_5 - R_1$
- $R_3 \to R_3 - 3R_2$
- $R_3 \leftrightarrow R_4$
- $R_3 \to \frac{1}{2}R_3, \quad R_4 \to \frac{1}{2}R_4, \quad R_5 \to 6\left(R_5 - \frac{1}{6}R_4\right)$
- $R_4 \to R_4 + R_5$
- $R_3 \to R_3 + \frac{1}{2}R_4 + \frac{1}{2}R_5$
- $R_1 \to R_1 + \frac{1}{6}R_4$

This reduces the system to the Reduced Row Echelon Form (RREF):

$$\begin{pmatrix} \mathbf{I} & \mathbf{y}^* \end{pmatrix} = \left(\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 0 & \frac{5}{12} \\ 0 & 1 & 0 & 0 & 0 & \frac{1}{5} \\ 0 & 0 & 1 & 0 & 0 & \frac{29}{20} \\ 0 & 0 & 0 & 1 & 0 & \frac{3}{2} \\ 0 & 0 & 0 & 0 & 1 & \frac{19}{10} \end{array}\right) \tag{2.2.12.11}$$

Multiplying back by the parameter x_4, the complete stoichiometric coefficient vector

is given by:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{pmatrix} = x_4 \begin{pmatrix} \frac{5}{12} \\ \frac{1}{5} \\ \frac{29}{20} \\ 1 \\ \frac{3}{2} \\ \frac{19}{10} \end{pmatrix} \tag{2.2.12.12}$$

which corresponds to the scalar relations:

$$x_1 = \frac{5}{12}x_4 \tag{2.2.12.13}$$
$$x_2 = \frac{1}{5}x_4 \tag{2.2.12.14}$$
$$x_3 = \frac{29}{20}x_4 \tag{2.2.12.15}$$
$$x_5 = \frac{3}{2}x_4 \tag{2.2.12.16}$$
$$x_6 = \frac{19}{10}x_4 \tag{2.2.12.17}$$

Evaluating with the sample numerical basis $x_4 = 5.0$:

$$x_1 = \frac{25}{12} \approx 2.0833 \tag{2.2.12.18}$$
$$x_2 = 1.0000 \tag{2.2.12.19}$$
$$x_3 = 7.2500 \tag{2.2.12.20}$$
$$x_4 = 5.0000 \tag{2.2.12.21}$$
$$x_5 = 7.5000 \tag{2.2.12.22}$$
$$x_6 = 9.5000 \tag{2.2.12.23}$$

The moles of oxygen consumed per mole of glucose consumed during fermentation is given by:

$$\frac{x_3}{x_1} = \frac{\frac{29}{20}x_4}{\frac{5}{12}x_4} = \frac{29}{20} \times \frac{12}{5} = \frac{87}{25} = 3.48 \tag{2.2.12.24}$$

or using the numerical evaluation:

$$\frac{x_3}{x_1} = \frac{7.2500}{2.0833} = 3.48 \tag{2.2.12.25}$$

Rounding off to two decimal places, the answer is 3.48. The implementation in SymPy to find solutions is as follows:

codes/btq58/58.py

2.2.13 Let $a_0 = 0$ and define $a_n = \frac{1}{2}(1 + a_{n-1})$ for all positive integers $n \geq 1$. The least value of n for which $|1 - a_n| < \frac{1}{2^{10}}$ is _____. (GATE BT 2025) **Solution:**

The given recurrence relation is:

$$a_n = \frac{1}{2}\left(1 + a_{n-1}\right), \quad n \geq 1, \quad a_0 = 0 \tag{2.2.13.1}$$

Let $x\left(n\right) = a_n u\left(n\right)$, where $u\left(n\right)$ is the discrete unit step function:

$$u\left(n\right) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{2.2.13.2}$$

Incrementing the recurrence index to $n \geq 0$:

$$a_{n+1} - \frac{1}{2}a_n = \frac{1}{2}, \quad n \geq 0 \tag{2.2.13.3}$$

Multiplying by $u\left(n\right)$ yields the causal difference equation:

$$x\left(n+1\right) - \frac{1}{2}x\left(n\right) = \frac{1}{2}u\left(n\right), \quad n \geq 0 \tag{2.2.13.4}$$

(Equivalently, shifting the index by one sample gives $x\left(n\right) - \frac{1}{2}x\left(n-1\right) = \frac{1}{2}u\left(n-1\right)$ for $n \geq 1$).
The unilateral (one-sided) Z-transform of $x\left(n\right)$ is defined as:

$$X^+\left(z\right) = \mathcal{Z}^+\left(x\left(n\right)\right) = \sum_{n=0}^{\infty} x\left(n\right) z^{-n} \tag{2.2.13.5}$$

Applying the unilateral Z-transform to the advanced sequence $x\left(n+1\right)$:

$$\mathcal{Z}^+\left(x\left(n+1\right)\right) = \sum_{n=0}^{\infty} x\left(n+1\right) z^{-n} \tag{2.2.13.6}$$

$$= z\sum_{n=0}^{\infty} x\left(n+1\right) z^{-(n+1)} \tag{2.2.13.7}$$

$$= z\left(\sum_{k=0}^{\infty} x\left(k\right) z^{-k} - x\left(0\right)\right) \tag{2.2.13.8}$$

$$= zX^+\left(z\right) - zx\left(0\right) \tag{2.2.13.9}$$

Taking the unilateral Z-transform on both sides of (**??**):

$$\left(zX^+\left(z\right) - zx\left(0\right)\right) - \frac{1}{2}X^+\left(z\right) = \frac{1}{2}\mathcal{Z}^+\left(u\left(n\right)\right) \tag{2.2.13.10}$$

$$\left(z - \frac{1}{2}\right)X^+\left(z\right) - zx\left(0\right) = \frac{\frac{1}{2}}{1 - z^{-1}} \tag{2.2.13.11}$$

Since $a_0 = 0$, the initial condition is $x\left(0\right) = a_0 u\left(0\right) = 0$:

$$\left(z - \frac{1}{2}\right)X^+\left(z\right) = \frac{\frac{1}{2}}{1 - z^{-1}} \tag{2.2.13.12}$$

Multiplying both sides by z^{-1}:

$$\left(1-\frac{1}{2}z^{-1}\right)X^{+}\left(z\right)=\frac{\frac{1}{2}z^{-1}}{1-z^{-1}} \tag{2.2.13.13}$$

Solving algebraically for $X^{+}\left(z\right)$:

$$X^{+}\left(z\right)=\frac{\frac{1}{2}z^{-1}}{\left(1-z^{-1}\right)\left(1-\frac{1}{2}z^{-1}\right)},\quad \left|z\right|>1 \tag{2.2.13.14}$$

Performing partial fraction decomposition with $w=z^{-1}$:

$$\frac{\frac{1}{2}w}{\left(1-w\right)\left(1-\frac{1}{2}w\right)}=\frac{C_1}{1-w}+\frac{C_2}{1-\frac{1}{2}w} \tag{2.2.13.15}$$

where the residues are evaluated as:

$$C_1=\left[\frac{\frac{1}{2}w}{1-\frac{1}{2}w}\right]_{w=1}=\frac{1/2}{1/2}=1 \tag{2.2.13.16}$$

$$C_2=\left[\frac{\frac{1}{2}w}{1-w}\right]_{w=2}=\frac{1}{1-2}=-1 \tag{2.2.13.17}$$

Substituting C_1 and C_2 back into $X^{+}\left(z\right)$:

$$X^{+}\left(z\right)=\frac{1}{1-z^{-1}}-\frac{1}{1-\frac{1}{2}z^{-1}} \tag{2.2.13.18}$$

Applying the inverse Z-transform using (**??**) and (**??**):

$$x\left(n\right)=\left[1-\left(\frac{1}{2}\right)^{n}\right]u\left(n\right)=1-2^{-n},\quad n\geq 0 \tag{2.2.13.19}$$

Thus, for all $n\geq 0$:

$$a_n=1-2^{-n} \tag{2.2.13.20}$$

Applying the stopping threshold condition $\left|1-a_n\right|<\frac{1}{2^{10}}$:

$$\left|1-\left(1-2^{-n}\right)\right|<\frac{1}{2^{10}} \tag{2.2.13.21}$$

$$2^{-n}<2^{-10} \tag{2.2.13.22}$$

$$2^{n}>2^{10}\implies n>10 \tag{2.2.13.23}$$

Since n must be an integer, the minimal integer satisfying the strict inequality is:

$$n=11 \tag{2.2.13.24}$$

The implementation in C to simulate the recurrence and verify the threshold is as follows:

codes/btq59/ztransform.c

2.2.14 Let $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & k & 0 \\ 3 & 0 & -1 \end{pmatrix}$. If the eigenvalues of A are -2, 1, and 2, then the value of k is _____. (BT 2025)

Solution: From (**??**),

$$1 + k + (-1) = (-2) + 1 + 2 \implies k = 1 \tag{2.2.14.1}$$

The following code verifies the above result.

codes/btq63/eigen.py